

PROGRAM-5: Demonstrate CI integration by running jenkins in a Docker container

Program Structure:

Dockerfile

docker-compose.yml

Steps 1: Check whether docker service is running

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p5$ docker version
Client: Docker Engine - Community
 Version:           29.1.2
 API version:       1.52
 Go version:        go1.25.5
 Git commit:        890dcca
 Built:             Tue Dec  2 21:55:19 2025
 OS/Arch:           linux/amd64
 Context:           default

Server: Docker Engine - Community
 Engine:
  Version:           29.1.2
  API version:       1.52 (minimum version 1.44)
  Go version:        go1.25.5
  Git commit:        de45c2a
  Built:             Tue Dec  2 21:55:19 2025
  OS/Arch:           linux/amd64
  Experimental:      false
 containerd:
  Version:           v2.2.0
  GitCommit:         1c4457e00facac03ce1d75f7b6777a7a851e5c41
 runc:
  Version:           1.3.4
  GitCommit:         v1.3.4-0-gd6d73eb8
 docker-init:
  Version:           0.19.0
  GitCommit:         de40ad0
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p5$ docker compose version
Docker Compose version v5.0.0
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p5$
```

Step 2: Create a Dockerfile

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p5$ nano Dockerfile
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p5$ cat Dockerfile
FROM jenkins/jenkins:latest

USER root

RUN apt-get update && apt-get install -y docker.io

RUN usermod -aG docker jenkins

USER jenkins
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p5$ █
```

Step 3: Create a docker-compose.yml

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p5$ nano docker-compose.yml
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p5$ cat docker-compose.yml
version: '3.8'

services:
  jenkins:
    build: .
    container_name: jenkins
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock

volumes:
  jenkins_home:
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p5$ █
```

Step 4: Build the docker image

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/Lab/p5$ sudo docker build -t program-5 .
[sudo] password for 1rv24mc017_ananya_ganapati_hegde:
[+] Building 398.4s (8/8) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 176B
=> [internal] load metadata for docker.io/jenkins/jenkins:latest
=> [auth] jenkins/jenkins:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/3] FROM docker.io/jenkins/jenkins:latest@sha256:d8ec54c56ad567ba18ddc3a1ef47ecdd27de8ff3d5fd4eeed2c2b10e63595d2e
=> => resolve docker.io/jenkins/jenkins:latest@sha256:d8ec54c56ad567ba18ddc3a1ef47ecdd27de8ff3d5fd4eeed2c2b10e63595d2e
=> => sha256:d8ec54c56ad567ba18ddc3a1ef47ecdd27de8ff3d5fd4eeed2c2b10e63595d2e 3.12kB / 3.12kB
=> => sha256:584c1da6223eb5a0da2f6f998ac9879684c828ad2d95bc6c643821b2777ac350 5.24MB / 5.24MB
=> => sha256:bc358ff4a912c11a9b777c6be84de53aeabc28c3022cebed79aa23fe4fbec620 2.58kB / 2.58kB
=> => sha256:53c88f1dfb79b2f207f7f1a03a45e0dc5ed208b9f496de16b98f81189dc0392 49.29MB / 49.29MB
=> => sha256:f2b6797c62d535ba07802aa757613eba9ba9d313967d5989a644e43e49e32c08 12.68kB / 12.68kB
=> => sha256:7a9125f681128d88c9f1895fe73effc28b180fccaf5dd33218bb6bdaf119b0d9 61.59MB / 61.59MB
=> => sha256:3a2e238b684ca0f15c29c3f42d881ed8c11e2e340506cec35c108ece61c2d930 1.23kB / 1.23kB
=> => sha256:b9c84d7526088df48af4d8bef3bb597a981abb1573a1b8c8ba4ec2647bf73a93 184B / 184B
=> => sha256:a3aaf4e0058df121d4b4fa9ffdb0bf07fc38a24ea0c2017e93e62e9c241f1fc75 97.07MB / 97.07MB
=> => extracting sha256:53c88f1dfb79b2f207f7f1a03a45e0dc5ed208b9f496de16b98f81189dc0392
=> => sha256:1821596f97db0cce91f065c4b4779be6912a6c73d8b0227ecddf6e0801d52a3 188B / 188B
=> => sha256:d54eb0e9facce73a941039c8008af0ebbc2289227bb2d21e599dc245cf09f5 6.29MB / 6.29MB
=> => extracting sha256:7a9125f681128d88c9f1895fe73effc28b180fccaf5dd33218bb6bdaf119b0d9
=> => sha256:7c31f64c1f331d7a0fc797e2145fd2ddec22e584dbde5647fcb5d0687cfe1d04 72.05MB / 72.05MB
=> => extracting sha256:584c1da6223eb5a0da2f6f998ac9879684c828ad2d95bc6c643821b2777ac350
=> => extracting sha256:3a2e238b684ca0f15c29c3f42d881ed8c11e2e340506cec35c108ece61c2d930
=> => extracting sha256:b9c84d7526088df48af4d8bef3bb597a981abb1573a1b8c8ba4ec2647bf73a93
=> => sha256:6037b00d1ddb635ff8065a7bac03c3ad6571d147474643e60b2fa5a5513 2.23kB / 2.23kB
=> => sha256:78a40a76459e1f89b17deb8f65055cc8a8f0c1d467e9231d8c9c1e07ed708764 1.29kB / 1.29kB
=> => sha256:cfd06ba8b2efa6b6af77b7434c9bbbf397d787798ec4d65a01d9ca0c1782670b 391B / 391B
=> => extracting sha256:a3aaf4e0058df121d4b4fa9ffdb0bf07fc38a24ea0c2017e93e62e9c241f1fc75
=> => extracting sha256:1821596f97db0cce91f065c4b4779be6912a6c73d8b0227ecddf6e0801d52a3
=> => extracting sha256:d54eb0e9facce73a941039c8008af0ebbc2289227bb2d21e599dc245cf09f5
=> => extracting sha256:7c31f64c1f331d7a0fc797e2145fd2ddec22e584dbde5647fcb5d0687cfe1d04
=> => extracting sha256:6037b00d1ddb635ff8065a7bac03c3ad6571d147474643e60b2fa5a5513
=> => extracting sha256:78a40a76459e1f89b17deb8f65055cc8a8f0c1d467e9231d8c9c1e07ed708764
=> => extracting sha256:cfd06ba8b2efa6b6af77b7434c9bbbf397d787798ec4d65a01d9ca0c1782670b
=> [2/3] RUN apt-get update && apt-get install -y docker.io
=> [3/3] RUN usermod -u 0 docker jenkins
=> exporting to image
=> => exporting layers
=> => writing image sha256:fbdf45402cf447ba00fe7465cba903d594a0e0d82056ff940b124f78530301bf
=> => naming to docker.io/library/program-5
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/Lab/p5$
```

Step 5: Run the docker image

```
1rv24nc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/Lab/p5$ sudo docker images
INFO → In Use
IMAGE                                ID                                DISK USAGE    CONTENT SIZE  EXTRA
gcr.io/k8s-minikube/kicbase:v0.0.48 c6b5532e987b    1.31GB        0B            U
innova_backend:latest                837eb2b98be0    191MB         0B            U
innova_frontend:latest               6e7ea2a836b6    56.2MB        0B            U
innova_nongodb:latest                7f9b8eeea2157   915MB         0B            U
kicbase/stable:v0.0.48               c6b5532e987b    1.31GB        0B            U
program-5:latest                     fbd45402cf4     925MB         0B            U
quay.io/coreos/etcd:v3.5.13          bef2e3b99162    57.8MB        0B            U
1rv24nc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/Lab/p5$ sudo docker run program-5
Running from: /usr/share/jenkins/jenkins.war
webroot: /var/jenkins_home/war
2026-01-05 12:10:14.563+0000 [id=1] INFO winstone.Logger#logInternal: Beginning extraction from war file
2026-01-05 12:10:15.119+0000 [id=1] WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-01-05 12:10:15.155+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: jetty-12.1.5; built: 2025-12-03T22:18:24.732Z; git: 4985d09d7b6801e792d7b73946cf7b66e4be25d6; jvm 21.0.9+10-LTS
2026-01-05 12:10:15.403+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2026-01-05 12:10:15.442+0000 [id=1] INFO o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2026-01-05 12:10:15.656+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins home directory: /var/jenkins_home found at: EnvVars.masterEnvVars.get("JENKINS_HOME")
2026-01-05 12:10:15.721+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@1dc76fa1(Jenkins v2.544.1, b=files:///var/jenkins_home/war, a=AVAILABLE, h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@5eed2d86(STARTED))
2026-01-05 12:10:15.729+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@74b4083a(HTTP/1.1, (http/1.1)){0.0.0.0:8080}
2026-01-05 12:10:15.729+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Started oejs.Server@4d654825(STARTING)[12.1.5,sto=0] @576ms
2026-01-05 12:10:15.730+0000 [id=43] INFO winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2026-01-05 12:10:15.802+0000 [id=42] INFO jenkins.model.Jenkins#<init>: Starting version 2.544
2026-01-05 12:10:15.853+0000 [id=51] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-01-05 12:10:15.862+0000 [id=72] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-01-05 12:10:16.341+0000 [id=53] INFO jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-01-05 12:10:16.344+0000 [id=62] INFO jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-01-05 12:10:16.345+0000 [id=62] INFO jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-01-05 12:10:16.426+0000 [id=58] INFO jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-01-05 12:10:16.426+0000 [id=58] INFO jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-01-05 12:10:16.426+0000 [id=61] INFO jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-01-05 12:10:16.428+0000 [id=61] INFO jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-01-05 12:10:16.445+0000 [id=86] INFO hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2026-01-05 12:10:16.670+0000 [id=51] INFO jenkins.install.SetupWizard#init:
[LF]>
[LF]> *****
[LF]> *****
[LF]> *****
[LF]>
[LF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[LF]> Please use the following password to proceed to installation:
[LF]>
```

Step 6: Verify whether jenkins is running by navigating to

<http://localhost:8080>

http://localhost:8080

Getting Started

Customize Jenkins

Plugins extend Jenkins with additional features to support many different needs.

Install suggested plugins

Install plugins the Jenkins community finds most useful.

Select plugins to install

Select and install plugins most suitable for your needs.

Step 7: Display the docker images and remove the image

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p5$ sudo docker rmi -f program-5:latest
Untagged: program-5:latest
Deleted: sha256:fbdf45402cf447ba00fe7465cba903d594a0e0d82056ff940b124f78530301bf
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p5$
```